Hybrid Multiscale Spatial–Spectral Transformer for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification constitutes a significant foundation for remote sensing analysis. Transformer architecture establishes long-range dependencies with a self-attention mechanism (SA), which exhibits advantages in HSI classification. However, most existing transformer-based methods are inadequate in exploring the multiscale properties of hybrid spatial and spectral information inherent in HSI data. To countermeasure this problem, this work investigates a hybrid multiscale spatial–spectral framework (HMSSF). It innovatively models global dependencies across multiple scales from both spatial and spectral domains, which allows for cooperatively capturing hybrid multiscale spatial and spectral characteristics for HSI classification. Technically, a spatial–spectral token generation (SSTG) module is first designed to generate the spatial tokens and spectral tokens. Then, a multiscale SA (MSSA) is developed to achieve multiscale attention modeling by constructing different dimensional attention heads per attention layer. This mechanism is adaptively integrated into both spatial and spectral branches for hybrid multiscale feature extraction. Furthermore, a spatial–spectral attention aggregation (SSAA) module is introduced to dynamically fuse the multiscale spatial and spectral features to enhance the classification robustness. Experimental results and analysis demonstrate that the proposed method outperforms the state-of-the-art methods on several public HSI datasets.**

***Index Terms*— Hyperspectral image (HSI) classification, multiscale self-attention, transformer network.**

I. Introduction

HYPERSPECTRAL images (HSI) are represented by hundreds of continuous spectral bands in the electromagnetic spectrum, encompassing the characteristics of abundant spectral information and high spatial correlation. With the rapid development of remote sensing technology, HSI performs significant advantages in various remote sensing scenarios, such as mineral mapping [1], [2], [3], military reconnaissance [4], [5], [6], and precision agriculture [7], [8], [9], [10]. As a fundamental procedure in applications, HSI classification focuses on pixel-by-pixel distinguishing and allocating categorical labels for ground objects, which has attracted increasing attention in the remote sensing community [11], [12], [13], [14].

Traditional research on HSI classification commonly draws upon machine learning techniques [15], [16], [17], [18]. Classically, Melgani and Bruzzone [19] explored the utilization of support vector machines (SVMs) for the classification of hyperspectral remote sensing images. Li et al. [20] introduced multinomial logistic regression to model the posterior class distributions and combined active learning strategy for HSI classification. Notably, the previous methods primarily concentrate on the spectral information while disregarding the spatial dimension. This deficiency potentially restricts the classification performance, particularly for objects that exhibit similar spectral characteristics but different spatial representations. With the development of imaging technology, several works endeavor to integrate complementary spatial contextual information with spectral features for precise HSI classification mapping. For instance, Fauvel et al. [21] developed an extended morphological profile (EMP) for extracting spatial contextual features of hyperspectral data, followed by the integration with spectral information for classification. Duan et al. [22] devised a geodesic-based sparse manifold hypergraph method (GSMH) to reveal the intricate manifold structure and multivariate relationship of samples in HSI, and the classification capability is enhanced by integrating the hypergraph embedding and sparse representation (SR). However, the feature extraction and data fitting capabilities of these traditional methods prove insufficient when confronting complex environmental conditions, which results in performance bottlenecks for HSI classification.

With the rapid development of deep learning, convolutional neural networks (CNNs) have exhibited substantial potential in computer vision tasks, including HSI classification [23], [24], [25], [26], [27], [28], [29]. Chen et al. [30] first introduced the concept of deep learning into HSI classification and designed stacked autoencoders to hierarchically extract the high-level spectral features of hyperspectral data. The classification is performed by merging the deep-learned spectral features with handcrafted spatial information. The

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competitive performance on standard hyperspectral datasets demonstrated the effectiveness of deep learning-based models for HSI classification. Subsequently, Yang et al. [31] developed a dual-branch architecture combining 1-D CNN and 2-D CNN for capturing finer spectral and spatial features of hyperspectral data, respectively. Benefiting from the advantage of the 3-D convolution paradigm with spatial–spectral joint feature extraction, Zhong et al. [32] devised a 3-D convolution-based spectral–spatial residual network (SSRN), which is capable of capturing deep spectral–spatial blocks directly from raw 3-D cubes without additional feature engineering. The residual connection and batch normalization in spectral and spatial blocks effectively alleviate the accuracy decline phenomenon. To mitigate computational complexity, Roy et al. [33] further proposed a hybrid spectral architecture (HybridSN) by integrating spectral–spatial 3-D CNN blocks and spatial 2-D CNN modules. The 3-D CNN enables the joint spatial–spectral feature representation from spectral bands, and the 2-D CNN captures abstract-level spatial features. Despite achieving promising classification performance compared to traditional approaches, these aforementioned methods are still consistently limited by fixed convolutional receptive fields, resulting in the loss of characteristic details.

Recently, several methods endeavor to integrate the multiscale and multilevel feature extraction strategy into CNN backbone for HSI classification [34], [35], [36]. For example, He et al. [37] constructed an end-to-end multiscale 3-D deep DNN (M3D-DCNN) based on multiscale 3-D convolution blocks, which simultaneously meet the multiscale and multiresolution requirements in spatial and spectral dimensions. Shang et al. [38] proposed an HSI classification algorithm based on multiscale cross-branch response and second-order channel attention (MCRSCA), in which the complementary spatial context information is cooperatively captured by multiple convolution kernels with different sizes. Zhu et al. [36] introduced a novel deformable CNN-based HSI classification network (DHCNet). The deformable convolutions are adaptively adjusted according to the complex spatial contexts of hyperspectral data, thereby benefiting from dynamic receptive fields. Additionally, Li et al. [39] presented a two-stream CNN backbone network to enhance the contextual representation capability. This architecture extracts both local and global spatial features simultaneously and incorporates a layer-specific regularization and normalization scheme to adaptively balance the contribution of the parallel features. As the mainstream backbone architecture, CNN-based methods exhibit formidable locally contextual modeling capability. However, they struggle to effectively establish long-range spectral correlations among sequence-based HSI bands.

Benefiting from the powerful long-distance sequence modeling capabilities, Transformer architecture has been adeptly investigated for HSI classification task [40], [41], [42], [43], [44], [45]. In particular, Hong et al. [46] reconsidered the HSI classification from a spectral sequential perspective and pioneeringly constructed a pure transformer-based SpectralFormer (SF) network. The SpectralFormer focuses on capturing the spectral sequence information between adjacent HSI bands, which yields groupwise spectral embeddings with significant classification superiority. To comprehensively integrate both spatial and spectral information, Sun et al. [47] presented a spectral–spatial feature tokenization transformer (SSFTT) for HSI classification. This architecture first extracts shallow spectral and spatial features through a hybrid convolutional module and then performs feature transformation through a Gaussian weighted token module for high-level spectral–spatial representation learning with a transformer encoder. Similarly, Roy et al. [41] developed a new morphological transformer (morphFormer) that implements a learnable spectral and spatial morphological network, where spectral and spatial morphological convolution operations are used (in conjunction with the attention mechanism) to improve the interaction between the structural and shape information of the HSI token and the CLS token. Zhao et al. [48] proposed a groupwise separable convolutional ViT (GSC-ViT), which incorporates the groupwise separable strategy into both the CNN and transformer, enabling the extraction of local and global spectral–spatial information. Furthermore, Zhong et al. [49] devised a novel spectral–spatial transformer network (SSTN), which breaks the long-range limitations of convolution kernels by integrating spatial attention and spectral association modules, and incorporates a factorized architecture search model to determine the layer-level operations and block-level orders. Additionally, Peng et al. [50] constructed a dual-branch spatial–spectral transformer with cross attention (CASST), where the spatial branch captures fine-grained spatial details and the spectral branch establishes interdependencies among spectral sequences. The incorporated spatial–spectral cross-attention module within transformer blocks further facilitates interactive feature fusion between the spatial and spectral branches. Although the above Transformer-based studies have exhibited impressive performance in HSI classification by modeling long-range global dependencies, they primarily concentrate on single-scale feature extraction while lacking the multiscale information of HSI data.

The multiscale extraction strategy enjoys the advantage of capturing representations across different scales without information distortion or redundancy, which has demonstrated significant effectiveness in various classification tasks [51], [52], [53], [54]. Recently, several researchers attempt to integrate the multiscale strategy with Transformer architecture to comprehensively explore discriminative features for HSI classification. Considering that spectral resolution at different scales can capture different levels of spectral features, He et al. [55] developed a cross-spectral vision transformer (CSiT) to extract pixelwise multiscale spectral features. CSiT effectively captures the correlation between spectral bands on different scales, however, ignoring the multiscale attribute in the spatial domain. Huang et al. [56] proposed a two-stage training-based Transformer SS-MTr for HSI classification, where a multiscale data augmentation strategy based on multiscale bilinear interpolation is introduced for data preprocessing. Nevertheless, these existing methods still perform inadequately in exploring the multiscale characterization of HSI data, which inevitably restricts the classification performance.

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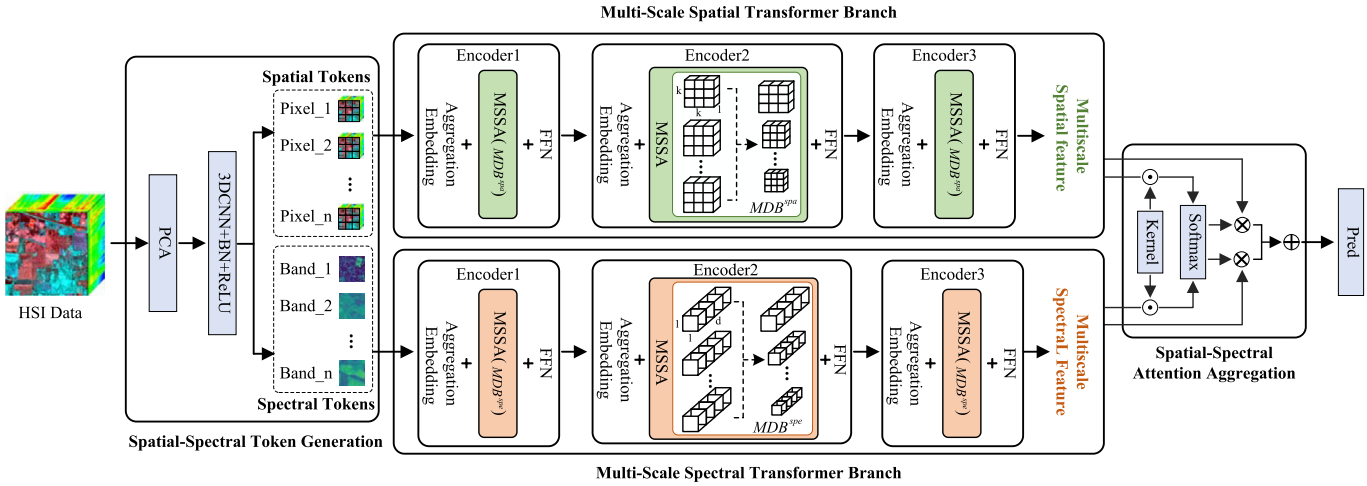


Fig. 1. Overall architecture of HMSSF, which consists of a spatial–spectral token generation module, a multiscale spatial transformer branch, a multiscale spectral transformer branch, and a spatial–spectral attention aggregation module.

To compensate aforementioned limitations, this work investigates a hybrid multiscale spatial–spectral framework (HMSSF). It innovatively models global dependencies across multiple scales from both spatial and spectral domains, which aims to capture the hybrid multiscale spatial and multiscale spectral features for HSI classification. Specifically, the developed HMSSF comprises two parallel spatial transformer branch and spectral transformer branch, which possess analogous structures hierarchically stacked by multiple encoders. To achieve the multiscale feature extraction, a multiscale downsample block (MDB)-based multiscale self-attention mechanism (MSSA) is integrated into the encoders. By constructing different dimensional keys and values for different attention heads with MDB, MSSA performs attention encoding at multiple scales per attention layer, which enables establishing the global dependencies of multiscale features. Notably, MDB is specifically designed with distinct downsampling operations for spatial and spectral branches. After parallel processing with dual branch, an attention-based spatial–spectral attention aggregation (SSAA) module is further introduced, which aims for adaptive fusion of multiscale spatial and spectral features by allocating dynamic learnable weights. The proposed HMSSF demonstrates superior performance in HSI classification compared with other state-of-the-art approaches. The main contributions are summarized as follows.

1) An HMSSF is proposed for HSI classification. By cooperatively taking advantage of hybrid multiscale spatial and spectral information, the classification capability can be significantly enhanced.
2) An MDB-based MSSA is introduced, which explicitly allows the self-attention heads within the same layer to account for different granularity features.
3) An SSAA module is designed for adaptive feature fusion. By adaptively assigning fusion weights for spatial and spectral features, the interaction of spatial and spectral information can be adequately explored.
4) Extensive experiments are verified on four public hyperspectral datasets: Indian Pines, Pavia University, Houston 2013, and Houston 2018. The results indicate the effectiveness and superiority of the proposed method.

The remaining sections of this article are organized as follows. Section II provides a comprehensive description of the proposed method. Section III outlines the experimental results and analyses. Conclusion and future work are discussed in Section IV

II. Proposed Network

The original hyperspectral data is expressed as a 3-D tensor $I \in \mathbb{R}^{H\times W\times C}$, with H and W representing the spatial dimensions, and C denoting the spectral dimension. Several hundred spectral bands encompass abundant discriminative spatial and spectral characteristics. However, the high dimensionality also suffers from the Hughes phenomenon, which compromises the prediction performance for HSI classification. To mitigate this issue, dimensionality reduction is initially performed on the original hyperspectral data through principal component analysis (PCA) [57]. As depicted in Fig. 1, the transformed image with PCA is represented as $I_{\text{PCA}} \in \mathbb{R}^{H\times W\times c}$, where c refers to the reduced spectral dimension. Considering that the adjacent pixels can supplement spatial information for the central pixel, the modified image is further partitioned into a sequence of 3-D overlapping patch cubes $\{x_i \in \mathbb{R}^{P\times P\times c}\}_{i=1}^{H\times W}$, and the category is determined by the central pixel of each patch.

A. Hybrid Multiscale Spatial–Spectral Framework

The architecture of the HMSSF is illustrated in Fig. 1, which consists of a spatial–spectral token generation (SSTG) module, a multiscale spatial transformer branch, a multiscale spectral transformer branch, and an SSAA module. Initially, the cropped patch cube is fed into SSTG module to acquire the spatial token and spectral token. Subsequently, the generated tokens are respectively input into the corresponding upper Multiscale Spatial Transformer Branch (MTB^{spa}) and the lower Multiscale Spectral Transformer Branch (MTB^{spe}), which performs the independent capture of multiscale spatial features and multiscale spectral representations, respectively.

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Ultimately, the extracted multiscale features are collaboratively integrated into SSAA module for adaptive fusion, and then sent into a classifier to achieve the final HSI classification.

Taking the cropped pixelwise patch cube $x \in \mathbb{R}^{P\times P\times c}$ as input, the SSTG$(\cdot)$ first maps the cube into the spatial token and spectral token. The tokenization process can be formulated as follows:

$$\left[\mathcal{T}_{\text{spatial}}, \mathcal{T}_{\text{spectral}}\right] = \text{SSTG}(x) \tag{1}$$

where $\mathcal{T}_{\text{spatial}} = \{z_1, z_2, \ldots, z_m\} \in \mathbb{R}^{1\times D_{\text{spa}}}$ represents the generated spatial token, and $\mathcal{T}_{\text{spectral}} = \{s_1, s_2, \ldots, s_n\} \in \mathbb{R}^{1\times D_{\text{spe}}}$ denotes the spectral token.

After complimenting the tokenization operation, the spatial token and spectral token are parallelly transmitted into the multiscale spatial transformer branch $\text{MTB}^{\text{spa}}(\cdot)$ and multiscale spectral transformer branch $\text{MTB}^{\text{spe}}(\cdot)$ to implement the independent multiscale spatial and spectral feature extraction. The procedure can be formulated as follows:

$$f_{\text{spatial}} = \text{MTB}^{\text{spa}}\left(\mathcal{T}_{\text{spatial}}\right) \tag{2}$$

$$f_{\text{spectral}} = \text{MTB}^{\text{spe}}\left(\mathcal{T}_{\text{spectral}}\right). \tag{3}$$

As depicted in Fig. 1, the upper MTB^{spa} branch and the lower MTB^{spe} branch exhibit similar architecture, which is hierarchically stacked by multiple transformer encoders. Each encoder consists of an aggregation embedding, MSSA, and feed-forward network (FFN). The complete structure of encoder is illustrated in Fig. 2. Let f be the encoder input, it first undergoes the aggregation embedding as follows:

$$f = \text{Linear}(\text{Aggr}(f, \varepsilon)) \tag{4}$$

where $\text{Aggr}(\cdot)$ denotes the aggregation operation with the scale ε. For the spatial branch, aggregation is performed along the spatial dimension, while for the spectral branch aggregation is performed along the spectral dimension.

Unlike traditional single-scale self-attention, the proposed MSSA establishes global dependency at different scales by integrating an MDB, which facilitates effective multiscale feature extraction. Notably, MDB performs different downsampling operations for the spatial branch and spectral branch. Taking f as input, the process can be formulated as follows:

$$Y = \begin{cases} f + \text{MSSA}\left(\text{Norm}(f), \text{MDB}^{\text{spa}}\right), & \text{for } \text{MTB}^{\text{spa}} \\ f + \text{MSSA}\left(\text{Norm}(f), \text{MDB}^{\text{spe}}\right), & \text{for } \text{MTB}^{\text{spe}}. \end{cases} \tag{5}$$

Here, $\text{MSSA}(\cdot, \text{MDB}^{\text{spa}})$ refers to the MSSA that integrates the spatial-oriented MDB, and $\text{MSSA}(\cdot, \text{MDB}^{\text{spe}})$ is specifically tailed for the spectral characteristic.

Then, the intermediate representation is further transmitted to the FFN$(\cdot)$ comprising two linear layers and an activation function, which enhances representation capability through mining nonlinear relationships

$$f^{\text{out}} = Y + \text{FFN}(\text{Norm}(Y)). \tag{6}$$

The LayerNorm and residual connection are added to alleviate overfitting and gradient vanishing.

After parallel processing with dual branch, the captured multiscale spatial feature and spectral feature are collaboratively integrated into the SSAA$(\cdot)$ for adaptive feature fusion.

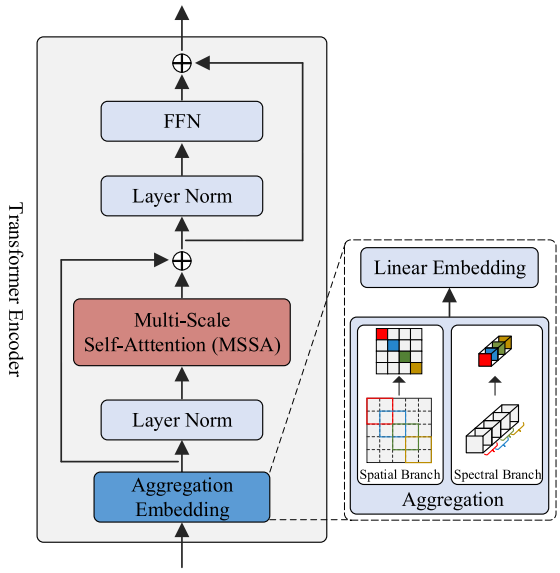


Fig. 2. Structure of Transformer encoder.

This step can be represented as follows:

$$f_{\text{spa-spe}} = \text{SSAA}\left(\text{AVG}\left(f_{\text{spatial}}\right), \text{AVG}\left(f_{\text{spectral}}\right)\right) \tag{7}$$

where $\text{AVG}(\cdot)$ denotes the average pooling operation.

Finally, the fused multiscale feature is sent into the classifier Classifier$(\cdot)$ comprised of multilayer perceptron layers to acquire the classification results

$$\text{pred} = \text{Classifier}\left(f_{\text{spa-spe}}\right). \tag{8}$$

Below, the implementation of SSTG module, MSSA module, and SSAA module will be described in detail.

B. Spatial–Spectral Token Generation

Traditional transformer-based methods typically exploit 2-D convolutional operation on the patches for HSI data tokenization, which inevitably distorts the spectral characteristics of the query pixels. In contrast, the presented SSTG module incorporates a 3-D convolution block and adds conversion operations to generate the spatial token and spectral token, which are subsequently transmitted into the corresponding multiscale transformer branch for feature extraction.

To be specific, SSTG takes HSI patch cube $x \in \mathbb{R}^{P\times P\times c}$ as input. The 3-D convolution block consists of a 3-D convolution layer 3D_Conv, a batch normalization layer BN, and a rectified linear activation function ReLU. The generated spatial–spectral token can be expressed as follows:

$$\mathcal{T}_{\text{spa-spe}} = \text{ReLU}(\text{BN}(3D_\text{Conv}(x))). \tag{9}$$

For the computational procedure of 3-D convolution, the generated value $v_{i,j}^{\nu,y,z}$ at position (ν, y, z) of the jth feature map in the ith layer can be expressed by

$$v_{i,j}^{\nu,y,z} = b_{i,j} + \sum_{\lambda}\sum_{\alpha=0}^{\eta_i-1}\sum_{\varphi=0}^{\delta_i-1}\sum_{\sigma=0}^{\gamma_i-1} \omega_{i,j,\lambda}^{\alpha,\varphi,\sigma} \times v_{i-1,\lambda}^{\nu+\alpha,\nu+\varphi,\nu+\sigma} \tag{10}$$

where $b_{i,j}$ represents the bias parameter, $\omega_{i,j,\tau}^{p,q,r}$ denotes the trainable kernel value at the position $(\alpha, \varphi, \sigma)$ in the λth

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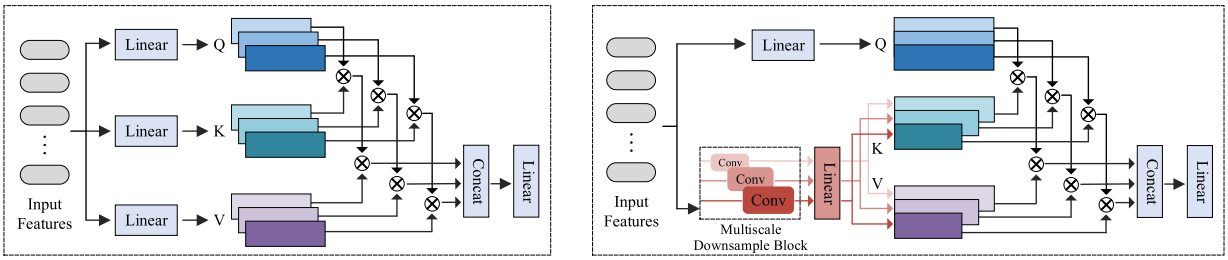


Fig. 3. Detail structures of (a) traditional self-attention mechanism and (b) MSSA.

feature map, and η_i, δ_i, and γ_i indicate the 3-D kernel size along the height, width, and depth dimension, respectively. The BN operation is to normalize the output of 3-D convolution to prevent the gradient explosion and gradient disappearance. The ReLU activation function is employed to enhance the feature nonlinearity.

To achieve spatial tokenization, SSTG subsequently separates $\mathcal{T}_{\text{spa-spe}} \in \mathbb{R}^{k\times k\times d}$ along the spatial dimension, which forms a set of 2-D spatial tokens as $\mathcal{T}_{\text{spatial}} = \{z_1, z_2, \ldots, z_d\} \in \mathbb{R}^{k\times k}$. The generated $\mathcal{T}_{\text{spatial}}$ is then flattened from the matrix structure into vector form, followed by an addition with a learnable positional encoding $\text{Pos}_{\text{spa}} \in {}^{1\times D_{\text{spa}}}$. The spatial tokenization process can be formulated as follows:

$$\mathcal{T}_{\text{spa}} = \text{Flatten}\big(\mathcal{T}_{\text{spa-spe}}, \text{axis} = \text{spatial}\big) + \text{Pos}_{\text{spa}} \tag{11}$$

where Flatten$(\cdot)$ encompasses the separation and flattening operations, and the vector length D_{spa} is equal to $k \times k$.

For the spectral sequence tokenization, $\mathcal{T}_{\text{spa-spe}} \in \mathbb{R}^{k\times k\times d}$ is partitioned with the spectral dimension to form the spectral tokens as $\mathcal{T}_{\text{spectral}} = \{s_1, s_2, \ldots, s_{k\times k}\} \in \mathbb{R}^{1\times d}$. After that, a spectral positional encoding $\text{Pos}_{\text{spe}} \in {}^{1\times D_{\text{spe}}}$ is added, and the entire tokenization process for the spectral branch is completed as follows:

$$\mathcal{T}_{\text{spe}} = \text{Flatten}\big(\mathcal{T}_{\text{spa-spe}}, \text{axis} = \text{spectral}\big) + \text{Pos}_{\text{spe}} \tag{12}$$

where D_{spe} equals d. The positional embedding is supplemented to retain the position information of patch tokens.

C. Multiscale Self-Attention Mechanism

Traditional self-attention mechanism models the long-range dependencies with single-scale attention heads, inherently restricting the receptive field of the attention layer. To mitigate this limitation, this work introduces an MSSA that captures different granularity information through multiple self-attention heads with different scales. MSSA is significantly integrated into both the spatial transformer branch and the spectral transformer branch to separately capture the multiscale spatial features and multiscale spectral features of hyperspectral data.

The traditional self-attention mechanism is illustrated in Fig. 3(a). Given the input token $\mathcal{T} \in \mathbb{R}^{N\times d}$, it is projected into Query ($Q \in \mathbb{R}^{N\times d}$), Key ($K \in \mathbb{R}^{N\times d}$), and Value ($V \in \mathbb{R}^{N\times d}$) matrices as follows:

$$(Q, K, V) = \big(\mathcal{T}W^Q, \mathcal{T}W^K, \mathcal{T}W^V\big) \tag{13}$$

where N and C represent the length and dimension of the input sequence, respectively, and W^Q, W^K, and W^V denote the learnable weight matrices of linear projection. For capturing informative features from different subspaces, Q, K, and V are generally split into h attention heads for parallel self-attention computation, denoted as follows:

$$\begin{aligned} Q &= \big\{Q_i \in {}^{N\times(d/h)}\big\}_{i=1}^{h} \\ K &= \big\{K_i \in {}^{N\times(d/h)}\big\}_{i=1}^{h} \\ V &= \big\{V_i \in {}^{N\times(d/h)}\big\}_{i=1}^{h}. \end{aligned} \tag{14}$$

Notably, the Q_i, K_i, and V_i across different heads within one attention layer maintain identical dimension of (d/h). Such uniform information granularity inevitably constrains the receptive field of the attention layer, resulting in coarse-grained and single-scale global dependency modeling. The traditional SA is inadequate for HSI classification tasks that exhibit multiscale characteristics.

Comparatively, the structure of the developed MSSA is depicted in Fig. 3(b). The key idea of MSSA is to render the dimension of K_i and V_i to be nonidentical across different heads within the same attention layer, thereby achieving attention modeling at multiple scales. Specifically, MSSA first transforms the input tokens into different scales through an MDB, and then performs linear projection on them to obtain inconsistent scale keys and values across different heads. This process can be formulated as follows:

$$\begin{aligned} Q_i &= \mathcal{T}W_i^Q \\ K_i &= \text{MDB}(\mathcal{T}, r_i)W_i^K \\ V_i &= \text{MDB}(\mathcal{T}, r_i)W_i^V \end{aligned} \tag{15}$$

where W_i^Q, W_i^K, and W_i^V represent the weight parameters of the linear projection in the ith attention head. The MDB$(\cdot, r_i)$ denotes the downsample operation in the ith head with the downsample scale r_i.

Subsequently, different attention heads in the attention layer are employed to model the attention of features at different scales. The attention output of each head is computed as a weighted sum of the values V_i. The correlation matrix is generated by performing point multiplication on the corresponding query vector Q_i and key vector K_i, followed by a softmax activation function. The computation procedure for each head is as follows:

$$\text{head}_i = \text{Attention}(Q_i, K_i, V_i) = \text{SoftMax}\left(\frac{Q_i K_i^{\text{T}}}{\sqrt{d_k}}\right)V_i \tag{16}$$

where $\sqrt{d_k}$ represents the contraction coefficient for gradient stabilization during backpropagation, and d_k is the dimension

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of K_i. Ultimately, the attention outputs of h heads are concatenated together and then mapped to generate the final output, which can be formulated as follows:

$$\mathrm{MSSA} = \mathrm{Linear}(\mathrm{Conact}(\mathrm{head}_i)), \quad i = 1, 2, \ldots, h \tag{17}$$

where Conact$(\cdot)$ represents the concatenation operation, and Linear$(\cdot)$ denotes the linear mapping layer.

Benefited from the powerful multiscale capacities, MSSA is significantly integrated into both the spatial transformer branch and the spectral transformer branch. As illustrated in Fig. 1, the MDB is separately designed based on the characteristics of spatial and spectral information.

Considering the planar structure of spatial information, MDB in the spatial branch is constructed with h 2-D convolution layers with different scales. The spatial token first reshapes into 2-D form and then goes through the convolution operation to implement the downsampling. The MDB for spatial branch can be reformulated as follows:

$$\mathrm{MDB}^{\mathrm{spa}} = \left\{2D_\mathrm{Conv}\left(\mathrm{Re\ shape}\left(X_{\mathrm{spatial}}\right), r_i\right)\right\}_{i=1}^{h}. \tag{18}$$

The downsample scale r_i is therefore instantiated as the kernel and stride size of the ith 2-D convolution.

For the spectral branch with stereoscopic property, MDB is composed of h 3-D convolution layers with different scales. The spectral tokens undergo reshaping along the spectral dimension and are subsequently subjected to the convolution operation to execute downsampling. The MDB for spectral branch can be reformulated as follows:

$$\mathrm{MDB}^{\mathrm{spe}} = \left\{3D_\mathrm{Conv}\left(\mathrm{Re\ shape}\left(X_{\mathrm{spectral}}\right), r_i\right)\right\}_{i=1}^{h}. \tag{19}$$

Similarly, r_i denotes the kernel and stride of the ith 3-D convolution, which varies across the temporal dimension while the height and width remain fixed at 1×1.

D. Spatial–Spectral Attention Aggregation

To comprehensively leverage hybrid spatial and spectral information, the generated multiscale spatial feature and spectral feature are further transmitted into the SSAA module, which performs adaptive feature fusion by allocating dynamic learnable weights. The specific procedure of SSAA is illustrated in Fig. 1.

Considering that the generated spatial feature $f_{\mathrm{spatial}} \in \mathbb{R}^{1\times D_{\mathrm{spa}}}$ and spectral feature $f_{\mathrm{spectral}} \in \mathbb{R}^{1\times D_{\mathrm{spe}}}$ from different branches may exhibit inconsistent dimensions, a linear layer Linear$(\cdot)$ is first introduced to transform the features into identical dimensions for subsequent weighted fusion

$$f_{\mathrm{spatial}} = \mathrm{Linear}\left(f_{\mathrm{spatial}}\right) \tag{20}$$

$$f_{\mathrm{spectral}} = \mathrm{Linear}\left(f_{\mathrm{spectral}}\right). \tag{21}$$

Taking both spatial and spectral features as $f = \{f_{\mathrm{spatial}}, f_{\mathrm{spectral}}\}$ input, the objective of SSAN is to learn the corresponding dynamic weights $w = \{w_{\mathrm{spatial}}, w_{\mathrm{spectral}}\}$ for adaptive feature aggregation. Inspired from the attention scheme, a kernel q is first learned to filter the feature vectors via dot product

$$u_i = q^{\mathrm{T}} f_i, \quad i = \{\mathrm{spatial}, \mathrm{spectral}\}. \tag{22}$$

TABLE I
NAME AND NUMBER OF SAMPLES OF EACH CLASS ON THE INDIAN PINES DATASETS WITH 10% LABELED DATA

Indian Pines dataset			
No.	Class Name	Train	Test
1	Alfalfa	5	41
2	Corn-notill	143	1285
3	Corn-mintill	83	747
4	Corn	24	213
5	Grass-pasture-mowed	48	435
6	Grass-trees	73	657
7	Grass-pasture	3	25
8	Hay-windrowed	48	430
9	Oats	2	18
10	Soybean-notill	97	875
11	Soybean-mintill	245	2210
12	Soybean-clean	59	534
13	Wheat	20	185
14	Woods	126	1139
15	Buildings	39	347
16	Stone	6	84
	Total	1024	9225

The filter generates a vector that represents the significance of the corresponding feature. To convert the significance to weights w_i subject to $\sum_i w_i = 1$, SSAN further passes u_i to a softmax operator and achieves all positive weights as follows:

$$w_i = \frac{e^{u_i}}{\sum_j e^{u_j}}, \quad i, j = \{\mathrm{spatial}, \mathrm{spectral}\}. \tag{23}$$

Afterward, the dynamic weights are multiplied with the corresponding features, followed by an elementwise addition operation. The aggregated feature can be formulated as follows:

$$f_{\mathrm{out}} = w_{\mathrm{spatial}} f_{\mathrm{spatial}} + w_{\mathrm{spectral}} f_{\mathrm{spectral}}. \tag{24}$$

As a result, SSNN effectively combines the multiscale features from both spatial and spectral domains, generating more comprehensive representations for HSI classification.

III. PERFORMANCE EVALUATION

A. Datasets Description

To demonstrate the classification capability of the proposed HMSSF, four publicly released HSI databases are provided for evaluation, including Indian Pines, Pavia University, Houston 2013, and Houston 2018. The details of the aforementioned datasets are described as follows.

1) Indian Pines: The dataset was collected by the Airborne/Visible Infrared Imaging Spectrometer (AVIRIS) imaging an Indian pine tree over Northwestern Indiana in 1992. The imaging wavelength of the spectrometer spans from 0.4 to 2.5 μm. After removing the water absorption channels, the dataset consists of 200 spectral bands and 145×145 pixels, with a spatial resolution of 20 m per pixel. There are a total of 10 249 ground object pixels, representing 16 distinct categories. Table I provides the detailed data division in the experiment.

2) Pavia University: The dataset was captured by the Reflective Optics System Imaging Spectrometer (ROSIS) over Pavia, Northern Italy. The imaging wavelength of the spectrometer ranges from 0.43 to 0.86 μm. After removing the noisy bands, the dataset encompasses 103 spectral bands and

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TABLE II
NAME AND NUMBER OF SAMPLES OF EACH CLASS ON THE PAVIA UNIVERSITY DATASETS WITH 5% LABELED DATA

Pavia University dataset			
No.	Class Name	Train	Test
1	Asphalt	332	6299
2	Meadows	932	17717
3	Gravel	105	1994
4	Trees	153	2911
5	Painted metal sheets	67	1278
6	Bare soil	251	4778
7	Bitumen	67	1263
8	Self-blocking bricks	184	3498
9	Shadows	47	900
	Total	2138	40638

TABLE III
NAME AND NUMBER OF SAMPLES OF EACH CLASS ON THE HOUSTON 2013 DATASETS WITH 10% LABELED DATA

Houston 2013 dataset			
No.	Class Name	Train	Test
1	Healthy grass	125	1238
2	Stressed grass	125	1241
3	Synthetic grass	70	690
4	Trees	124	1231
5	Soil	124	1229
6	Water	33	321
7	Residential	127	1255
8	Commercial	124	1231
9	Road	125	1239
10	Highway	123	1214
11	Railway	123	1222
12	Parking Lot 1	123	1220
13	Parking Lot 2	47	464
14	Tennis Court	43	423
15	Running Track	66	653
	Total	1502	14871

610 × 340 pixels, with a spatial resolution of 1.3 m per pixel. There are a total of 42 776 ground sample points, categorized into nine types including Asphalt, Gravel, Trees, and so on. The division details for training and testing sets are described in Table II.

3) Houston 2013: The dataset was imaged by the ITRES CASI-1500 sensor over the University of Houston campus and its surrounding areas, provided by the 2013 GRSS Data Fusion Contest. It includes 144 spectral bands ranging in wavelength from 0.38 to 1.05 μm, and consists of 340 × 1905 pixels with a spatial resolution of 2.5 m per pixel. There are a total of 16 373 sample pixels, covering 15 challenging land cover categories. The dataset is split into training data and testing data, as displayed in Table III.

4) Houston 2018: The dataset was acquired by the National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and its neighboring areas. It spans a spectral range of 0.38–1.05 μm with 48 bands at a 1-m ground sampling distance (GSD). The image comprises 601 × 2384 pixels. There are a total of 504 856 sample points with 20 categories. The precise numbers of training and testing samples are exhibited in Table IV.

TABLE IV
NAME AND NUMBER OF SAMPLES OF EACH CLASS ON THE HOUSTON 2018 DATASETS WITH 5% LABELED DATA

Houston 2018 dataset			
No.	Class Name	Train	Test
1	Healthy grass	595	9204
2	Stressed grass	1768	30734
3	Synthetic grass	36	648
4	Evergreen trees	685	12910
5	Deciduous trees	252	4769
6	Soil	226	4290
7	Water	14	252
8	Residential buildings	2001	37771
9	Non-residential buildings	11196	212556
10	Roads	2359	43507
11	Sidewalks	2060	31969
12	Crosswalks	78	1440
13	Major thoroughfares	2365	43983
14	Highways	494	9371
15	Railways	348	6589
16	Paved parking lots	576	10924
17	Unpaved parking lots	8	138
18	Cars	354	6193
19	Trains	268	5101
20	Stadium seats	342	6482
	Total	26025	478831

B. Experimental Settings

1) Evaluation Metrics: Following the state-of-the-art methods in HSI classification, overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (Kappa) are adopted as the evaluation metrics. To guarantee fairness in comparison, all experiments are conducted under identical experimental conditions, and the reported results are derived as the averages from ten consecutive experiments.

2) Competitive Approaches: To validate the effectiveness of the proposed HMSSF, three kinds of representative HSI classification approaches are selected for comparison, including conventional methods, CNN-based methods, and Transformer-based methods. The network architecture and experimental parameters are consistent with their respective original papers.

a) *Conventional Methods:* Two classical representative conventional methods are involved including SVM [19] and SuperPCA [17]. The SVM processes the spectral features of HSI data, which is implemented through a Gaussian radial basis function (RBF) kernel function. For the SuperPCA, PCA is employed within each superpixel block to facilitate feature extraction in homogeneous patches.

b) *CNN-based Methods:* Two representative CNN-based methods are selected including SSRN [32] and HybridSN [33]. The SSRN comprises four iterative spatial and spectral residual blocks, each consisting of 3-D convolutional layers, BN, and ReLU activation functions. The patch size is set to 7 × 7. For HybridSN, it is a hybrid CNN network that integrates three 2-D and three 3-D convolution layers to extract both spatial and spectral features for classification. The patch size is 15 × 15 with 100 epochs for all the datasets.

c) *Transformer-based Methods:* Eight Transformer-based methods are compared including SF [46], SSFTT [47], MorphFormer [41], GSC-ViT [48], SSTN [49], CASST [50], SS-MTr [56], and DBCTNet [58]. The SF focuses on spectral sequence attributes, which consist of four transformer encoders with each incorporating

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TABLE V
SENSITIVITY ANALYSIS FOR THE PROPOSED METHOD WITH DIFFERENT SCALES OF 3D_CONV KERNEL

3D_Conv	Indian Pines			Pavia University			Houston 2013			Houston 2018		
Kernel Scales	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
3 & [3, 4, 4]	98.87	98.14	98.72	99.74	99.53	99.65	99.27	99.31	99.21	96.41	93.27	95.33
3 & [3, 5, 5]	98.88	97.99	98.72	99.69	99.52	99.59	99.26	99.34	99.2	96.42	92.13	95.21
5 & [3, 3, 3]	**98.97**	**98.21**	**98.82**	**99.84**	**99.67**	**99.79**	99.37	99.40	99.32	**96.84**	**94.10**	**95.88**
5 & [5, 4, 4]	98.92	98.15	98.76	99.76	99.56	99.68	**99.44**	**99.48**	**99.4**	96.83	93.29	95.87
7 & [5, 5, 5]	98.90	98.18	98.75	99.76	99.61	99.69	99.41	99.44	99.37	96.63	92.56	95.62

a cross-layer adaptive fusion module. The SSFTT takes advantage of hybrid CNN and Transformer, which is implemented with a 2-D convolution layer, a 3-D convolution layer, and a transformer encoder for feature extraction. The MorphFormer consists of a convolutional layer, a grouped separable convolutional layer, a self-attention module based on morphology joint in spectral–spatial domain, and a linear layer for classification. The GSC-ViT integrates the groupwise separable convolution module and groupwise separable multihead self-attention module, utilizing the same input size of 8×8 as MorphFormer. The SSTN exploits factorization architecture search (FAS) to determine the hierarchical operation and block-level order within the network. The CASST constructs a dual-branch spatial–spectral transformer with two spatial–spectral cross-attention encoders for classification. The input patch size is determined as 11×11 for both SSTN and CASST methods. Unlike the above methods focusing on single scale, the SS-MTr introduces a multiscale data augmentation module for data preprocessing, with the patch size determined as 17×17. The DBCTNet constructs a multiscale spectral feature extraction module, comprising four channel split convolution layers with respective kernel size (1,1,3), (1,1,7), (1,1,11), and (1,1,15) to enrich spectral signals.

3) Implementation Details: All the experiments are implemented on the PyTorch platform with one RTX 3090Ti GPU. The training epochs and batch size are set as 100 and 64, respectively. The Adam gradient descent optimizer with learning rate 0.001 is exploited for parameter optimization. Several hyperparameters in HMSSF are predefined. The PCA dimension for reduction is set to 30. The Classifier$(\cdot)$ consists of one perceptron layer with the embedding dimension of 240, mapping the fused multiscale spatial–spectral features into category results.

C. Parameter Analysis

In this section, a series of experiments are conducted to analyze and determine the optimal parameters for HMSSF. These parameters include the input patch sizes, the token scales, the embedding dimensions of encoders, and the multiscale settings.

1) Different Input Patch Sizes: Fig. 4 illustrates the classification capability of different input patch sizes (corresponding to the parameter P) on the four HSI datasets. The input patch ranges from 11×11 to 19×19 with a growth interval of 2.

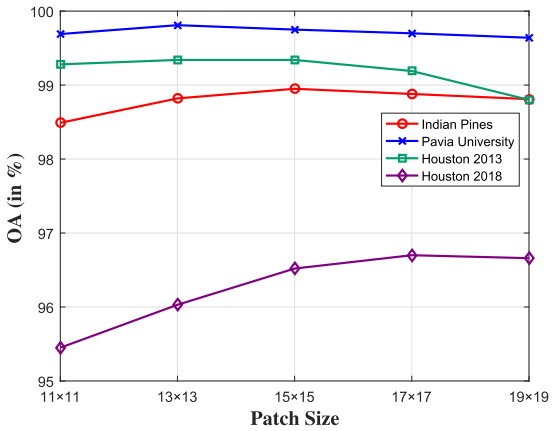


Fig. 4. Sensitivity analysis for the proposed method with different sizes of input patches.

Based on the results, the four datasets exhibit a similar variation tendency, where the performance gradually improves as the patch size increases. Upon reaching the optimal input size, the performance gradually declines. The possible reason might be that the larger patches may introduce redundant noise factors, leading to information aliasing. Consequently, a patch size of 15×15 is identified as suitable for the Indian Pines, Pavia University, and Houston 2013 datasets, and 17×17 for the Houston 2018 dataset.

2) Different 3D_Conv Kernel Scales: The spatial and spectral token scales are determined by the quantity and size of 3D_Conv kernels within the SSTG module. Table V depicts the classification accuracies achieved with various 3-D kernels on the four datasets. From the results, appropriately increasing the scale of 3D_Conv kernels contributes to more stable and rising outcomes. Accordingly, the 3-D kernel is established as 5 & [3, 3, 3] for the Indian Pines, Pavia University, and Houston 2018 dataset, and 5 & [5, 4, 4] for the Houston 2013 datasets.

3) Different Aggregation Scales: This section investigates the impact of different aggregation scales (corresponding to ε in (4)). Fig. 5 demonstrates the classification variations on the Indian Pines dataset, where the aggregation scales for the spatial branch set to {[2, 1], [2, 2], [3, 1], [4, 1], [5, 1]} and {[1, 1], [2, 2], [4, 1], [4, 2], [8, 1]} for the spectral branch. Taking $\varepsilon = [2, 1]$ as an example, the aggregation kernel is set to 2 and the aggregation stride is 1. From the results, it can be observed that the optimal aggregation scale for encoders 2 and 3 is [4, 1] for the spatial branch and [4, 2] for the spectral branch. For encoder 1, the aggregation scale remains at [1, 1] for both branches to preserve the scale of spectral

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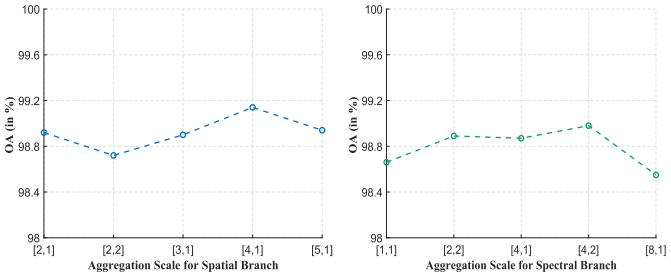


Fig. 5. Sensitivity analysis with different aggregation scales for spatial and spectral branches on the Indian Pines dataset. (a) Spatial branch and (b) spectral branch.

TABLE VI
SENSITIVITY ANALYSIS OF THE PROPOSED METHOD AND COMPARISON OF DIFFERENT EMBEDDING DIMENSIONS

Data	Stage	OAs (in%) of Different Dimensions				
		30	60	120	240	480
Indian Pines	Encoder 1	98.71	98.85	**98.96**	98.57	98.36
	Encoder 2	98.96	98.82	**99.03**	98.93	98.79
	Encoder 3	99.03	99.07	99.08	99.05	**99.09**
Pavia University	Encoder 1	99.81	**99.81**	99.74	99.7	99.63
	Encoder 2	99.81	**99.83**	99.81	99.79	99.74
	Encoder 3	99.83	99.79	**99.84**	99.79	99.80
Houston 2013	Encoder 1	98.93	**99.42**	99.38	99.36	99.31
	Encoder 2	99.42	99.40	**99.48**	99.40	99.33
	Encoder 3	99.37	**99.45**	99.42	99.35	99.33
Houston 2018	Encoder 1	96.36	**96.39**	95.98	95.52	95.34
	Encoder 2	96.39	96.46	**96.58**	96.56	96.53
	Encoder 3	96.58	**96.72**	96.55	96.22	96.29

and spatial tokens. These parameters are suitable for all the datasets.

4) Different Embedding Dimensions: Table VI demonstrates the classification sensitivity of HMSSF by comparing different embedding dimensions per encoder. The experiments are conducted across four datasets and the range of dimensions is set to $\{30, 60, 120, 240, 480\}$. The process of determining the optimal dimensions follows a low-to-high determination strategy. Taking the Indian Pines dataset as an example, the optimal dimension of the lower encoder 1 is first explored as 120, and then the dimension of encoder 2 is determined as 120. With the fixed dimensions of encoders 1 and 2, the impact of dimension for higher encoder 3 is further analyzed as 480. From the results, the optimal dimensions for the remaining Pavia University, Hoston 2013, and Hoston 2018 datasets are determined as $\{60, 60, 120\}$, $\{60, 120, 60\}$, and $\{60, 120, 60\}$, respectively. Notably, the embedding dimensions of the spectral branch and spatial branch remain consistent.

5) Different Multiscale Patterns: As the core of MSSA, MDB performs downsample operations with different scales on different heads within the same attention layer to achieve the multiscale attention objective. To determine the optimal multiscale pattern, we carry out mixed experiments on the number of attention heads and downsample scales (corresponding to h and r_i, respectively). Table VII provides six different multiscale patterns, where the number of heads ranges from 3 to 5 and the dowmsampe scale is formalized as Spa_scale & Spe_scale. Taking Pattern 2 as an example, the number of attention heads per attention layer is set to 3, and the corresponding dowmsampe scale (convulution kernel size) in encoder 1 is set as $\{1, 3, 6\}$ for the spatial branch and $\{1, 15, 30\}$ for the spectral branch. Subsequently, the downsample scale gradually decreases as the encoder deepens. Fig. 6 illustrates the classification accuracy variations with different patterns on the four HSI datasets. As observed from the figures, the different multiscale patterns exhibit slight fluctuations on the classification accuracy. In the case of the Indian Pines and Pavia University datasets, the Pattern 1 scheme achieves the highest performance. The most suitable multiscale scheme for the Houston 2013 and Houston 2018 datasets are Pattern 4 and Pattern 6, respectively.

D. Experimental Comparison With Competitive Approaches

To demonstrate the classification capabilities of the proposed HMSSF, extensive comparisons are performed with the competitive approaches in terms of classification accuracies and visualization maps. The quantitative results OA (%), AA (%), and Kappa (%) on the Indian Pines, Pavia University, Houston 2013, and Houston 2018 datasets are summarized in Tables VIII–XI, with the best results highlighted in bold. Corresponding visualization maps are provided in Figs. 7–10.

1) Indian Pines Dataset: The classification experiments on the Indian Pines dataset are conducted with 10% of the reference data. Table VIII provides the quantitative comparison results with each competitive approach. Conventional methods including SVM and SuperPCA exhibit limitations in processing HSI data with complex contextual information, consequently resulting in suboptimal classification performance. CNN-based methods such as SSRN and HyBridSN demonstrate relatively promising results; however, they struggle to capture global information due to their limited receptive fields. In contrast to Transformer-based methods, the proposed HMSSF also achieves relatively stable and superior performance across all categories. The classical SF method only focuses on modeling global dependencies of spectral information, while ignoring the exploration of spatial distribution. The improved CASST, SSTN, and SSFTT methods implement the joint extraction of both spectral and spatial characteristics without considering the multiscale nature of HSI data, inevitably suppressing their classification capabilities. The MorphFormer combines spectral and spatial morphological convolution with transformer, enhancing the interaction between image tokens and class tokens on structure and shape information in the HSI. The GSC-ViT incorporates the group-wise separable strategy into both the CNN and transformer, enabling the extraction of local and global spectral–spatial information. Comparatively, the proposed HMSSF still outperforms the aforementioned approaches, attributed to its capability of capturing multiscale spectral–spatial features. Recent SS-MTr and DBCTNet attempt to integrate the multiscale idea into the spatial domain, but compared with the proposed HMSSF, the dual-branch spatial and spectral modeling process based on MSSA ensures more comprehensive information acquisition. Additionally, the SSAA module further facilitates the adaptive interaction of spatial and spectral information, resulting in the best classification performance

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TABLE VII
DETAILS OF SIX MULTISCALE PATTERNS FOR THE PROPOSED METHOD

	Pattens	Encoder 1	Encoder 2	Encoder 3
Heads = 3	**Patten 1**	[1, 2, 3] & [1, 5, 10]	[1, 2, 3] & [1, 5, 10]	[1, 2, 3] & [1, 5, 10]
	Patten 2	[1, 3, 6] & [1, 15, 30]	[1, 2, 4] & [1, 10, 20]	[1, 2, 3] & [1, 5, 10]
Heads = 4	**Patten 3**	[1, 2, 3, 4] & [1, 5, 10, 15]	[1, 2, 3, 4] & [1, 5, 10, 15]	[1, 2, 3, 4] & [1, 5, 10, 15]
	Patten 4	[1, 3, 5, 7] &[1, 10, 20, 30]	[1, 2, 4, 5] & [1, 5, 15, 20]	[1, 2, 3, 4] & [1, 5, 10, 15]
Heads = 5	**Patten 5**	[1, 2, 3, 4, 5] & [1, 5, 10, 15, 20	[1, 2, 3, 4, 5] & [1, 5, 10, 15, 20	[1, 2, 3, 4, 5] & [1, 5, 10, 15, 20
	Patten 6	[1, 3, 5, 7, 9] & [1, 10, 20, 30, 40]	[1, 2, 4, 6, 7] & [1, 10, 15, 25, 30]	[1, 2, 3, 4, 5] & [1, 5, 10, 15, 20]

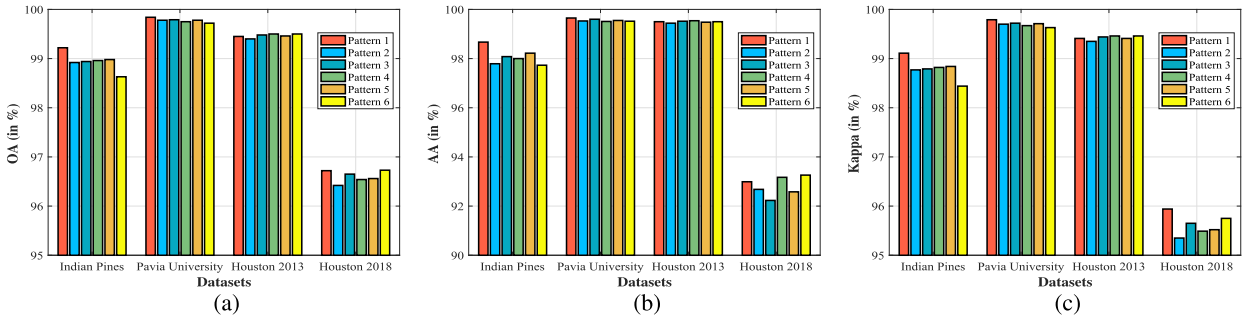


Fig. 6. Sensitivity analysis for the proposed method with different multiscale patterns in terms of (a) OA, (b) AA, and (c) κ.

TABLE VIII
CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASSES FOR THE INDIAN PINES DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods		CNN-based Methods		Transformer-based Methods								
	SVM	SuperPCA	SSRN	Hybrid-SN	SF	SSFTT	MorphFormer	GSC-ViT	SSTN	CASST	SS-MTr	DBCTNet	**HMSSF**
1	44.08	97.56	**100.0**	84.78	90.24	99.46	90.24	53.66	76.07	78.26	**100.0**	86.96	**100.0**
2	62.24	97.74	94.78	91.95	85.21	96.16	98.05	**98.37**	93.63	97.34	94.19	96.08	97.20
3	71.57	96.26	99.70	97.95	88.89	99.35	99.73	99.33	97.35	97.71	98.80	98.43	**100.0**
4	59.12	97.97	93.68	96.20	70.42	99.84	86.85	97.18	78.06	99.58	92.41	**100.0**	99.53
5	77.54	99.31	96.55	97.93	95.40	99.41	97.47	99.31	99.17	92.96	98.34	97.31	**99.54**
6	90.23	98.06	99.17	98.49	97.56	99.72	99.85	**100.0**	98.63	97.81	**100.0**	98.76	99.85
7	59.76	96.00	**100.0**	**100.0**	96.00	**100.0**	88.00	**100.0**	96.43	96.43	**100.0**	14.28	**100.0**
8	96.78	**100.0**	99.23	**100.0**	99.77	99.95	**100.0**	**100.0**	**100.0**	99.79	**100.0**	**100.0**	**100.0**
9	55.03	**100.0**	0.00	85.00	33.33	90.74	88.89	83.33	65.00	65.00	85.00	0.00	88.89
10	70.92	96.34	**100.0**	97.53	93.37	98.69	99.09	98.97	97.84	92.80	99.59	98.56	98.63
11	69.55	98.29	99.48	98.66	93.03	99.69	98.87	98.37	98.04	98.37	99.59	99.47	**99.82**
12	58.19	98.29	**98.56**	95.45	80.90	96.96	95.32	96.25	88.53	91.40	94.10	94.77	98.31
13	92.35	**100.0**	**100.0**	98.05	94.59	99.82	98.92	**100.0**	98.54	91.71	98.54	97.07	**100.0**
14	88.01	**100.0**	99.11	99.37	94.56	99.95	99.47	99.91	99.84	99.76	99.92	**100.0**	99.91
15	72.75	**99.71**	99.67	98.96	95.10	98.69	91.93	90.78	99.48	95.34	93.01	99.22	99.42
16	97.06	98.81	97.22	**100.0**	90.48	92.73	98.81	95.24	88.17	81.72	88.17	93.55	97.62
OA (%)	73.75	98.27	98.43	97.35	91.32	98.84	98.15	98.28	96.56	96.65	97.92	97.91	**99.22**
AA (%)	72.43	97.95	92.3	95.27	87.43	98.28	95.72	94.42	92.18	92.25	96.35	95.90	**98.67**
Kappa	70.28	98.02	98.2	96.97	90.11	98.60	97.89	98.04	96.07	96.18	97.63	97.62	**99.11**

for HMSSF. The accuracy improvements compared to the suboptimal SS-MTr reach 1.3% for OA, 2.32% for AA, and 1.48% for Kappa, respectively. These comparative results significantly demonstrate the effectiveness of the proposed method.

Additionally, the corresponding classification maps obtained by different approaches are visualized in Fig. 7. As can be observed, conventional methods such as SVM generally demonstrate noticeable noises owing to the limitations of feature extraction ability. The CNN-based and Transformer-based methods perform smoother and sharper, but still suffer from some detailed misclassifications. In contrast, the proposed HMSSF achieves the most consistent results with the ground truth, which presents the clearest category boundaries with minimal noise.

2) Pavia University Dataset: The experiments on the Pavia University dataset are performed with 5% of the reference samples. As can be observed from the classification results in Table IX, the proposed HMSSF outperforms the other competitive approaches in several categories. Compared to the suboptimal DBCTNet, the quantitative improvements in terms of OA, AA, and Kappa reach up to 0.15%, 0.37%, and 0.2%, respectively. Fig. 8 provides the classification visualization results of various methods, where the proposed HMSSF produces slighter edge information obfuscation. These phenomena further reveal the potentiality of integrating the

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TABLE IX

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASSES FOR THE PAVIA UNIVERSITY DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods		CNN-based Methods		Transformer-based Methods								
	SVM	SuperPCA	SSRN	Hybrid-SN	SF	SSFTT	MorphFormer	GSC-ViT	SSTN	CASST	SS-MTr	DBCTNet	**HMSSF**
1	85.17	98.90	99.90	99.14	94.75	99.70	99.78	99.63	99.52	99.17	99.32	**100.0**	**100.0**
2	91.34	99.22	99.89	99.86	98.14	99.90	99.95	99.90	**99.98**	99.95	99.90	99.97	99.93
3	80.23	**99.90**	98.83	97.48	84.70	98.95	94.93	97.32	92.04	98.28	97.57	99.38	99.00
4	92.07	97.39	**100.0**	98.48	96.81	98.52	98.52	98.29	99.35	93.08	96.54	97.64	99.48
5	99.45	99.77	**100.0**	99.93	99.22	99.53	**100.0**	99.89	**100.0**	98.66	99.78	99.70	99.92
6	88.12	99.71	99.94	99.74	94.37	**100.0**	**100.0**	99.74	99.24	99.60	**100.0**	**100.0**	**100.0**
7	86.29	98.02	**100.0**	99.59	84.56	**100.0**	99.29	99.59	99.85	99.62	99.32	**100.0**	**100.0**
8	83.05	97.36	95.55	98.10	92.65	98.43	98.60	98.63	94.95	94.73	92.12	99.78	99.91
9	**100.0**	99.78	99.89	96.79	**100.0**	97.89	98.56	98.93	99.79	96.41	93.35	97.10	98.67
OA (%)	88.71	98.96	99.48	99.29	96.60	99.55	99.41	99.46	98.95	98.63	98.63	99.69	**99.84**
AA (%)	89.33	98.66	99.33	98.79	93.91	99.21	98.85	99.10	98.30	97.72	97.55	99.28	**99.65**
Kappa	86.99	98.63	99.31	99.06	94.17	99.41	99.22	99.28	98.60	98.19	98.19	99.59	**99.79**

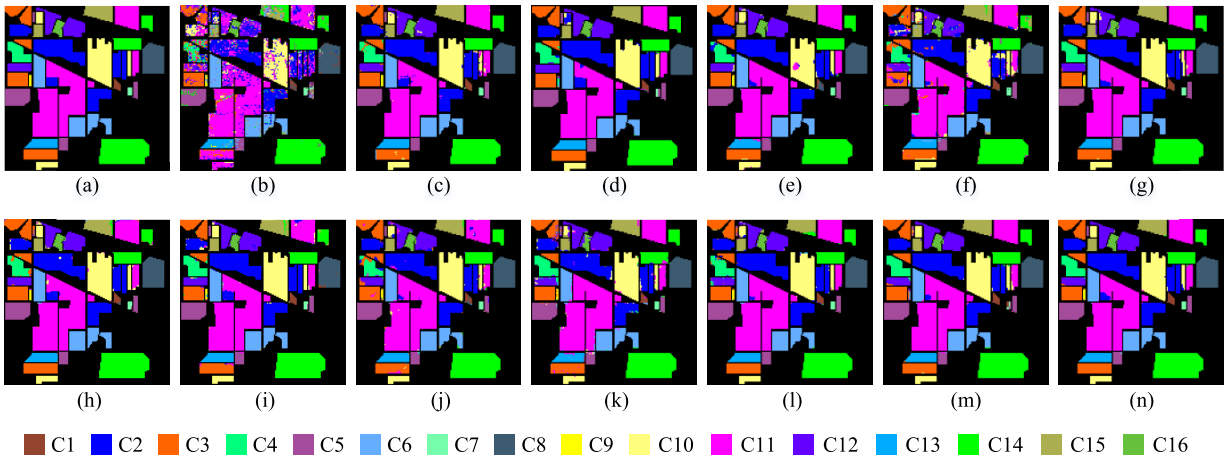


Fig. 7. Classification maps obtained by the compared methods on the Indian Pines dataset. (a) Reference map. (b) SVM. (c) SuperPCA. (d) SSRN. (e) HybridSN. (f) SF. (g) SSFTT. (h) MorphFormer. (i) GSC-ViT. (j) SSTN. (k) CASST. (l) SS-MTr. (m) DBCTNet. (n) HMSSF.

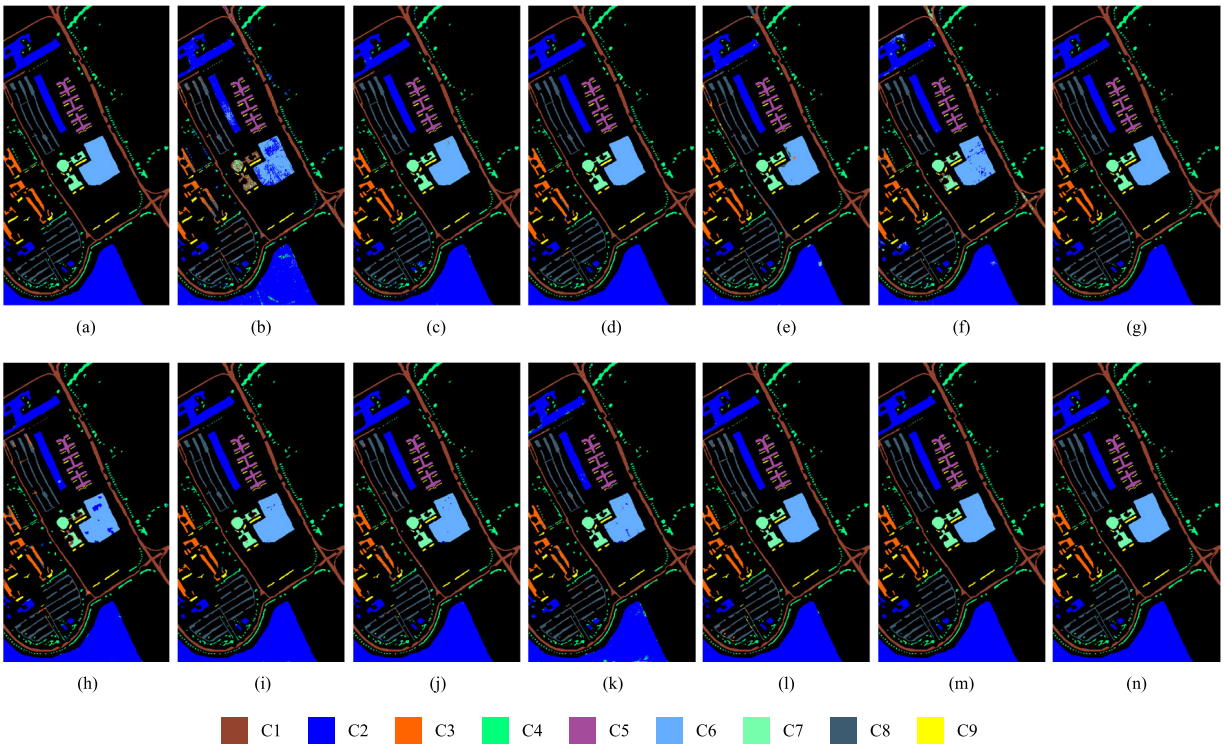


Fig. 8. Classification maps obtained by the compared methods on the Pavia University dataset. (a) Reference map. (b) SVM. (c) SuperPCA. (d) SSRN. (e) Hybrid-SN. (f) SF. (g) SSFTT. (h) MorphFormer. (i) GSC-ViT. (j) SSTN. (k) CASST. (l) SS-MTr. (m) DBCTNet. (n) HMSSF.

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TABLE X

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASSES FOR THE HOUSTON 2013 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods		CNN-based Methods		Transformer-based Methods								
	SVM	SuperPCA	SSRN	Hybrid-SN	SF	SSFTT	MorphFormer	GSC-ViT	SSTN	CASST	SS-MTr	DBCTNet	**HMSSF**
1	96.05	98.58	**99.91**	99.04	99.38	96.09	98.93	98.67	99.76	97.04	98.80	99.76	99.64
2	96.32	96.78	99.30	99.52	97.70	99.56	99.65	99.73	99.68	98.01	99.60	99.76	**99.91**
3	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	99.84	99.84	99.68	**100.0**	98.71	**100.0**	**100.0**	**100.0**
4	97.27	90.33	**100.0**	99.20	97.59	98.75	98.12	98.84	99.84	99.04	99.44	**100.0**	99.91
5	97.49	98.69	**100.0**	**100.0**	**100.0**	99.82	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
6	**100.0**	95.68	**100.0**	**100.0**	95.55	**100.0**	97.95	**100.0**	96.00	95.38	98.15	**100.0**	**100.0**
7	80.44	86.97	**99.48**	94.79	96.32	89.13	97.92	95.79	96.92	95.90	98.10	98.26	98.25
8	82.17	92.43	**100.0**	96.30	91.96	95.09	95.89	97.59	97.67	96.22	96.70	97.67	97.77
9	68.58	93.29	98.41	97.52	93.97	90.77	95.30	91.66	97.36	95.53	98.80	98.72	**99.47**
10	77.46	98.13	98.51	99.10	98.55	95.65	99.82	99.18	99.51	99.27	99.84	**100.0**	**100.0**
11	75.21	98.14	99.53	98.95	94.96	92.90	97.12	99.01	99.67	97.89	98.79	**100.0**	**100.0**
12	71.43	96.41	82.33	99.03	90.99	98.11	98.02	98.02	**99.68**	98.54	97.89	99.59	99.64
13	78.59	74.19	98.81	92.54	78.91	87.44	**99.29**	97.39	82.83	97.23	88.81	97.65	98.58
14	94.67	**100.0**	**100.0**	98.60	**100.0**	98.70	**100.0**	99.74	99.53	**100.0**	**100.0**	**100.0**	**100.0**
15	98.56	99.58	**100.0**	**100.0**	**100.0**	**100.0**	99.49	**100.0**	**100.0**	99.39	**100.0**	**100.0**	**100.0**
OA (%)	85.6	94.90	97.83	98.35	96.05	95.89	98.29	98.11	98.54	97.85	98.62	99.41	**99.50**
AA (%)	87.27	94.61	98.42	98.31	95.73	96.12	98.48	98.35	97.89	97.88	98.34	99.43	**99.54**
Kappa	84.13	94.49	97.65	98.22	95.73	95.56	98.15	97.95	98.42	97.68	98.50	99.36	**99.46**

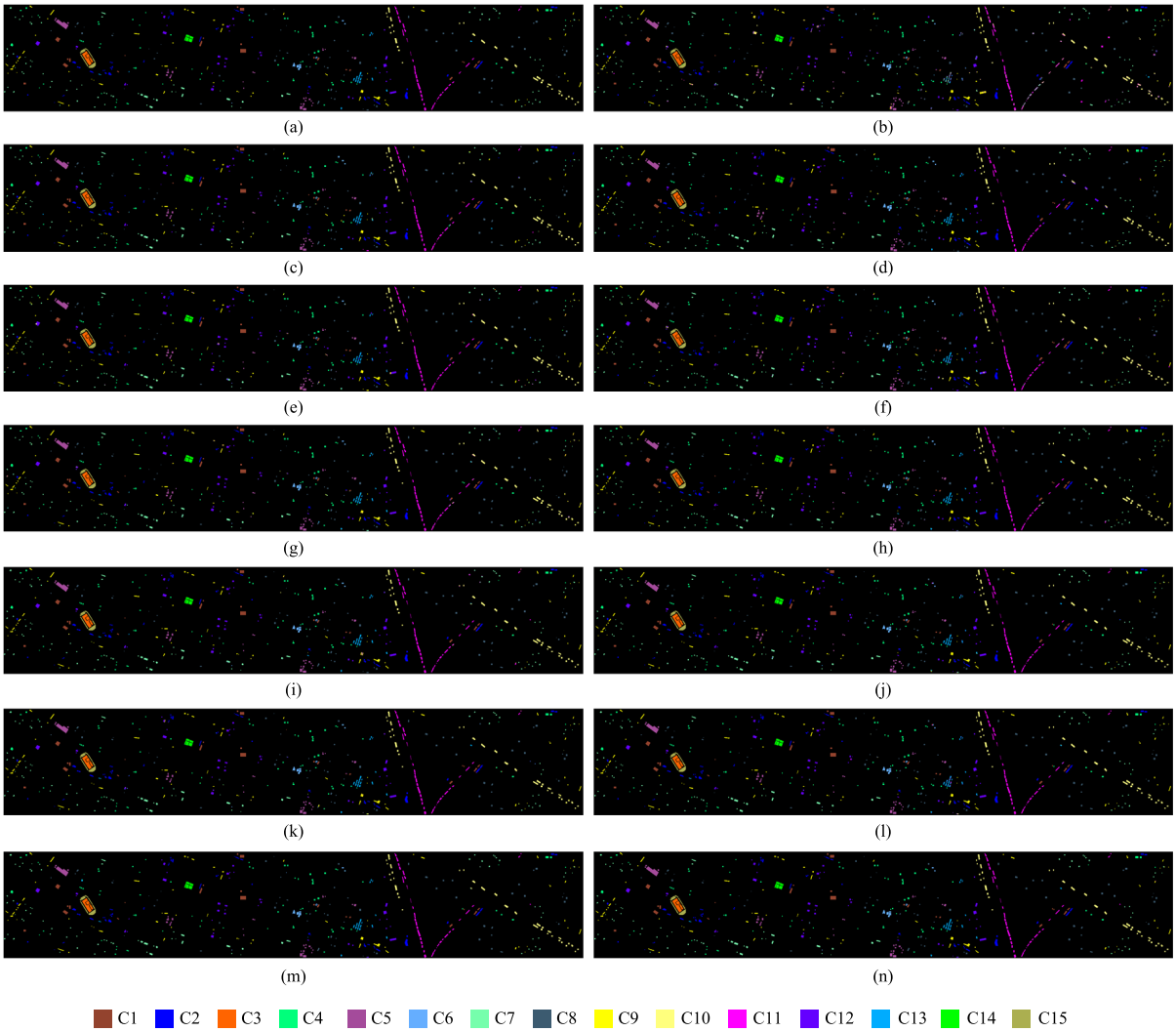


Fig. 9. Classification maps obtained by the compared methods on the Houston 2013 dataset. (a) Reference map. (b) SVM. (c) SuperPCA. (d) SSRN. (e) Hybrid-SN. (f) SF. (g) SSFTT. (h) MorphFormer. (i) GSC-ViT. (j) SSTN. (k) CASST. (l) SS-MTr. (m) DBCTNet. (n) HMSSF.

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TABLE XI

CLASSIFICATION ACCURACIES OF THE COMPARED METHODS IN TERMS OF OA, AA, κ, AND THE ACCURACIES OF EACH CLASSES FOR THE HOUSTON 2018 DATASET. THE BEST ACCURACIES ARE PRESENTED IN BOLD

Class	Conventional Methods		CNN-based Methods		Transformer-based Methods								
	SVM	SuperPCA	SSRN	Hybrid-SN	SF	SSFTT	MorphFormer	GSC-ViT	SSTN	CASST	SS-MTr	DBCTNet	**HMSSF**
1	92.19	74.14	**94.70**	90.16	89.81	74.67	85.83	85.97	87.99	79.41	85.09	79.97	87.82
2	89.46	92.53	94.01	92.81	96.22	94.68	96.07	96.20	92.53	**97.62**	93.81	95.72	93.29
3	70.24	99.70	0.00	99.56	99.54	98.31	99.85	99.35	98.98	**100.0**	99.85	97.66	97.54
4	90.29	96.49	94.17	98.21	97.50	96.43	98.16	95.85	96.90	97.90	**98.65**	96.15	96.93
5	80.47	89.03	92.81	92.65	91.43	85.70	93.08	89.44	88.69	**94.18**	93.89	91.97	87.84
6	76.49	98.42	97.14	99.98	99.49	99.72	99.91	99.54	99.78	99.78	**100.0**	99.51	99.95
7	**100.0**	99.62	0.00	64.66	67.19	73.52	94.47	99.17	93.98	94.36	93.61	98.87	94.86
8	65.71	96.93	96.25	97.64	94.95	94.12	97.09	96.13	96.01	98.35	97.80	97.13	**99.41**
9	78.64	97.70	98.77	99.26	97.55	98.59	97.27	98.39	98.84	**99.41**	98.98	98.68	99.29
10	51.22	89.81	82.03	91.25	86.68	84.28	82.97	85.72	83.02	**92.64**	88.09	84.73	90.34
11	59.37	80.00	87.07	78.97	72.80	84.62	78.71	79.61	77.32	86.18	77.47	81.03	**90.06**
12	0.00	23.59	9.97	28.85	26.56	31.76	3.68	19.05	25.89	25.96	25.16	10.74	**40.08**
13	54.18	96.68	90.97	93.33	93.11	89.17	94.70	93.80	90.62	94.22	94.76	91.03	**97.82**
14	76.29	96.88	80.79	99.41	95.80	94.27	93.74	93.70	94.93	**99.51**	96.97	96.21	98.53
15	59.83	99.85	98.35	99.74	99.82	98.63	99.58	99.29	98.86	99.78	**99.91**	98.34	99.83
16	78.02	96.50	94.91	98.89	95.69	95.50	98.29	96.60	96.08	98.67	98.34	95.11	**99.39**
17	0.00	90.21	0.00	93.84	94.96	74.82	0.00	96.40	73.29	**100.0**	92.47	93.84	96.40
18	44.49	**97.99**	91.12	97.05	92.04	92.70	94.65	94.29	91.57	95.92	95.07	94.47	96.30
19	80.19	97.34	98.93	99.14	97.98	96.24	99.29	98.37	96.74	98.88	98.44	97.82	**99.63**
20	90.18	**100.0**	95.80	**100.0**	99.74	99.48	99.41	98.69	99.50	99.91	99.99	99.38	**100.0**
OA (%)	73.01	94.53	94.11	95.56	93.69	93.77	93.87	94.44	93.74	96.47	95.10	94.26	**96.73**
AA (%)	66.86	90.67	74.89	90.77	89.44	87.86	85.34	90.78	89.08	92.63	91.42	89.92	**93.26**
Kappa	71.46	92.9	92.35	94.22	91.80	91.90	92.06	92.77	91.84	95.41	93.63	92.53	**95.75**

multiscale attention mechanism with hybrid spatial–spectral information.

3) Houston 2013 Dataset: The experiments on the Houston 2013 dataset are executed with 10% of the labeled samples. The quantitative classification accuracies are reported in Table X. From the results, the proposed HMSSF yields the most competitive performance in comparison with other studied methods, and exhibits surprising improvements in some categories, such as stressed grass, road, highway, and railway. Due to insufficient consideration of spatial information and multiscale strategies, typical Transformer-based SF architecture inevitably suffers from undesirable outcomes. Comparatively, the proposed HMSSF benefits from the hybrid spatial–spectral contextual information from the multiscale perspective, which exhibits a significant improvement of 3.45% in overall accuracy (OA), 3.81% in average accuracy (AA), and 3.73% in the Kappa coefficient. Regarding the visualization maps in Fig. 9, HMSSF is capable of producing the most precise classification details in this challenging scenario, which is distributed with discrete and local objects. These excellent improvements reveal the effectiveness and superiority of the proposed method.

4) Houston 2018 Dataset: The experiments on the Houston 2018 dataset are completed with 5% of the reference samples. Table XI lists the precise classification accuracies and Fig. 10 displays the corresponding maps. Similar conclusions can be drawn that HMSSF still demonstrates distinct advantages in quantitative performance and visual properties, which can be attributed to its excellent hybrid multiscale spatial–spectral information extraction ability. Drawing from the quantitative results, the proposed HMSSF achieves metrics of 96.73%, 93.26%, and 95.75% for OA, AA, and Kappa, respectively, which increased by 0.26%, 0.63%, and 0.34% compared with the suboptimal CASST. Accordingly, more accurate classification visualizations with fewer misclassifications can be observed in Fig. 10, particularly in areas such as major thoroughfares and nonresidential buildings, further validating the superiority of our method.

E. Robustness Assessment

To demonstrate the robustness of the proposed HMSSF, extensive experiments are conducted considering various proportions of training samples. Specifically, the selected percentage for the Indian Pines and Houston 2013 datasets covers an interval of $\{1.0\%, 2.0\%, \ldots, 10.0\%\}$, and for the Pavia University and Houston 2018 datasets is set to $\{0.5\%, 1.0\%, \ldots, 5.0\%\}$. Fig. 11 displays the performance variations with different percentages on the four HSI datasets, with the proposed HMSSF highlighted by red curves. The results indicate that the classification accuracy of HMSSF steadily increases with the percentages of training samples, which exhibit excellent robustness. Furthermore, HMSSF almost consistently outperforms other competitive methods across arbitrary percentages on the four testing datasets. These notable advantages further demonstrate the feasibility and superiority of the proposed method.

F. Computational Analysis

This section investigates the computational cost of the proposed HMSSF, focusing on model parameters, training time, and testing time. Table XIII illustrates the results with

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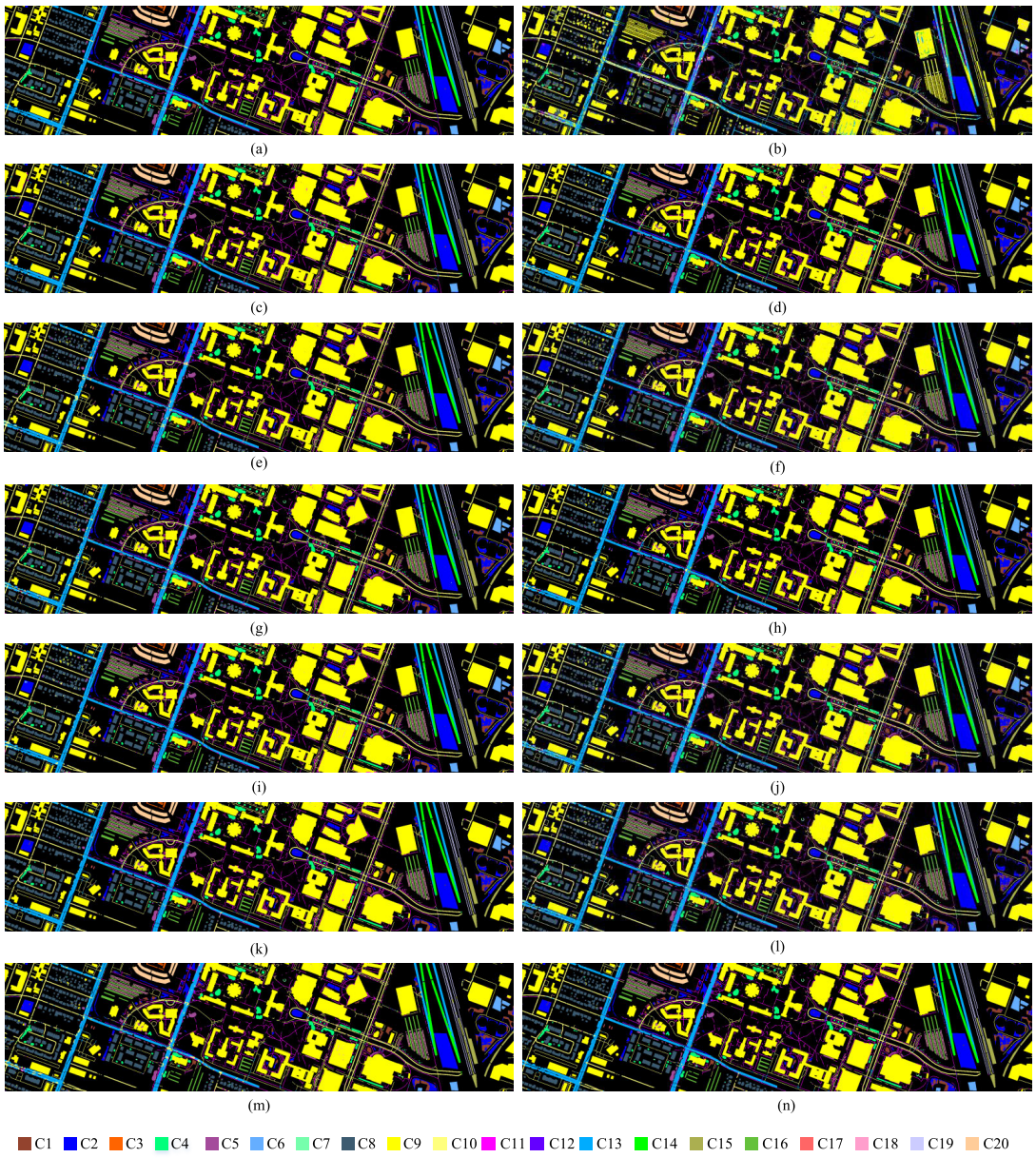


Fig. 10. Classification maps obtained by the compared methods on the Houston 2018 dataset. (a) Reference map. (b) SVM. (c) SuperPCA. (d) SSRN. (e) Hybrid-SN. (f) SF. (g) SSFTT. (h) MorphFormer. (i) GSC-ViT. (j) SSTN. (k) CASST. (l) SS-MTr. (m) DBCTNet. (n) HMSSF.

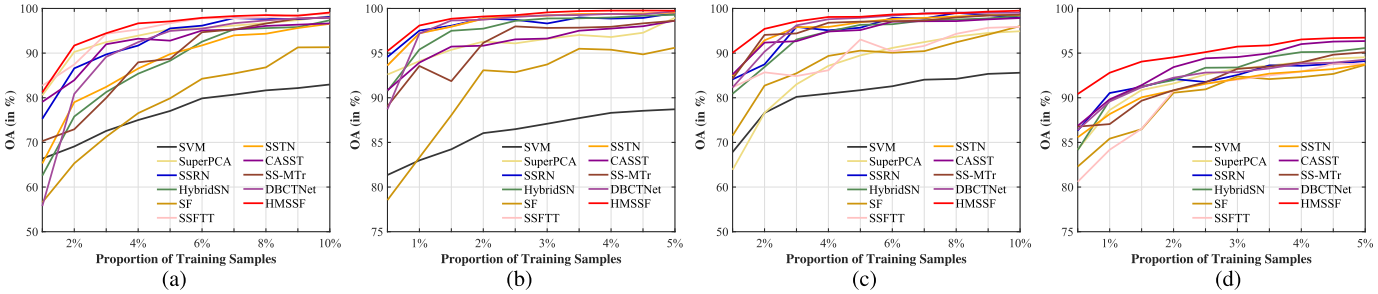


Fig. 11. Classification performance of the compared methods with different numbers of labeled data on the four datasets. (a) Indian Pines. (b) Pavia University. (c) Houston 2013. (d) Houston 2018.

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TABLE XII
ABLATION STUDY ON THE ACCURACY METRICS FOR THE HYBRID STRUCTURE AND MSSA

Spatial Branch	Spectral Branch	MSSA	Indian Pines			Pavia University			Houston 2013			Houston 2018		
			OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
	✓		89.9	83.17	86.52	92.08	88.2	89.43	88.2	88.22	87.24	93.65	88.31	91.70
	✓	✓	91.01	87.88	89.74	98.27	96.96	97.71	92.08	88.2	89.43	94.40	90.60	92.73
✓			96.37	93.39	95.84	99.52	98.91	99.36	98.88	99.00	98.78	96.20	93.17	95.06
✓		✓	98.89	97.43	98.73	99.64	99.24	99.42	99.13	99.23	99.06	96.35	93.56	95.33
✓	✓		98.58	96.64	98.38	99.56	99.11	99.42	99.06	99.18	98.99	96.18	90.59	95.03
✓	✓	✓	**99.22**	**98.67**	**99.11**	**99.65**	**99.79**	**99.50**	**99.50**	**99.54**	**99.46**	**96.73**	**93.26**	**95.75**

TABLE XIII
COMPUTATION COST ANALYSIS OF VARIOUS COMPARISON APPROACHES IN TERMS OF PARAMETERS (M), TRAINING TIME (S), AND TESTING TIME (S) ON FOUR DATASETS

		Hybrid-SN	SF	SSFTT	MorphFormer	GSC-ViT	SSTN	CASST	SS-MTr	DBCTNet	**HMSSF**
Indian Pines	Params	3.0583	0.1226	0.1484	0.1996	0.5636	0.3615	6.5275	1.6153	0.0061	5.6803
	Training time	20.58	244.29	62.24	147.31	92.33	49.23	89.69	61.5	36.94	183.66
	Inference time	0.41	2.38	0.46	3.37	1.64	0.63	3.23	1.24	0.51	4.42
Pavia University	Params	0.9141	0.1222	0.1254	0.1489	0.0778	0.3577	6.5275	1.4622	0.0059	0.5654
	Training time	37.67	637.29	50.26	219.29	152.26	95.30	182.67	124.19	74.08	371.87
	Inference time	1.108	6.38	1.48	27.2	13.256	2.63	13.4	4.99	2.07	19.7
Houston 2013	Params	0.1276	0.1133	0.1484	0.1547	0.0887	0.3609	6.5264	1.3667	0.0058	1.434
	Training time	26.58	284.74	7.52	183.91	116.68	68.82	129.34	86.22	51.28	292.80
	Inference time	0.32	1.70	0.52	92.03	45.41	0.93	4.76	1.57	0.81	7.05
Houston 2018	Params	0.9146	0.0999	0.1487	0.0798	0.065	0.3639	6.5316	1.314	0.0058	0.5527
	Training time	471.62	5828.94	560.15	2227.08	1610.01	1207.51	2164.33	1412.95	846.94	4979.06
	Inference time	16.20	28.16	23.54	178.30	84.95	33.65	161.34	48.69	24.90	261.06

other comparison methods on four dataset. Compared to CASST which is also equipped with similar dual-branch structure, the proposed HMSSF achieves better classification performance with significantly fewer computational parameters. This reduction might be attributed to our proposed MSSA and feature aggregation operations. Additionally, our method exhibits relatively longer training and testing times in modeling spatial and spectral information. The integration of 3-D convolutional layers and large-scale feature embeddings potentially increases the computational burden during the running process. In summary, the proposed HMSSF improves the classification performance significantly with an acceptable computational complexity. In our future work, we will consider exploring lightweight hyperspectral classification frameworks to reduce computational costs, achieving more efficient model execution and robust performance.

G. Ablation Study

1) Effectiveness of Hybrid Spatial–Spectral Structure in HMSSF: To underscore the detailed effectiveness of hybrid spatial–spectral structure in HMSSF, ablation experiments with single-branch setup by removing either the spectral or spatial branch are conducted on four HSI datasets. The classification results in terms of accuracy metrics are illustrated in Table XII. It can be observed that the performance of HMSSF based on hybrid spatial–spectral features outperforms both single spectral and single spatial branches, demonstrating strong robustness. Specifically, HMSSF surpasses the spectral branch by an average of 7.81% in OA, 10.84% in AA, and 9.73% in Kappa, respectively. Similarly, it exceeds the spatial branch by margins of 1.03%, 1.70%, and 1.20% for OA, AA, and Kappa, respectively. These disparities significantly accentuate the superiority of the hybrid structure in the original HMSSF, which is capable of adequately taking advantage of rich spectral and spatial semantic information inherent in HSI data.

2) Effectiveness of MSSA: This section comprehensively explores the rationality and significance of the MSSA module. Table XII presents the classification performance results. Specifically, the MSSA module is integrated into the spatial branch, spectral branch, and hybrid structures to substitute the traditional self-attention mechanism for ablation comparison. Compared to the MSSA focusing on multiscale feature extraction, the branches without the MSSA module perform the single-scale capture processes. As can be observed, the proposed MSSA benefits the discriminative feature extraction across three branch structures, exhibiting significant performance increases on all four datasets. This phenomenon experimentally demonstrates the efficacy of incorporating multiscale information for HSI classification.

3) Effectiveness of SSAA: To analyze the contribution of the SSAA module, ablation study is implemented with two common feature fusion strategies, including direct averaging (AVG) and concatenation (Concat). The corresponding classification comparisons are presented in Fig. 12. As an adaptive fusion module based on the attention concept, SSAA achieves dynamic fusion of multiscale spectral and spatial features by allocating learnable weights. Comparatively, the AVG and Concat strategies perform direct feature averaging or concatenation operations without considering spectral and spatial interaction. From the results, HMSSF incorporated with the SSAA module showcases higher performance compared to the AVG and Concat, demonstrating that the designed SSAA module definitely benefits the classification capability enhancement.

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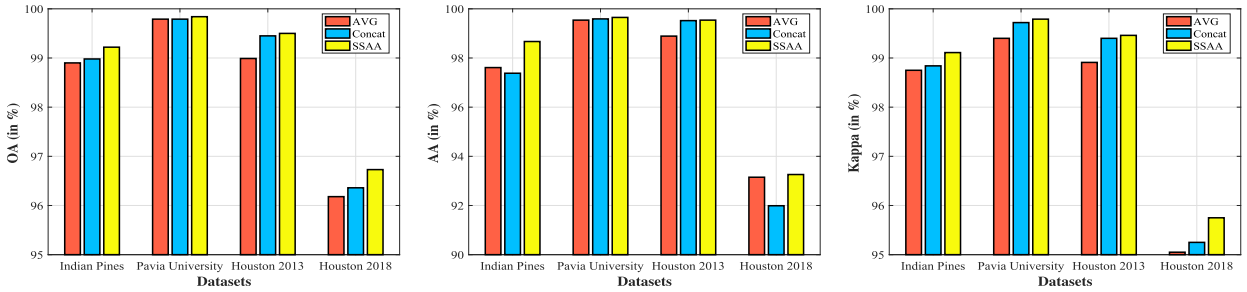


Fig. 12. Ablation study on the accuracy metrics in terms of (a) OA, (b) AA, and (c) κ for the SSAA.

IV. Conclusion

In this article, we propose an HMSSF, which innovatively models global dependencies across multiple scales from both spatial and spectral domains. By combining the MSSA, the proposed hybrid Transformer network has the ability to capture the multiscale spatial and spectral semantic characteristics. Benefiting from the SSAA module, the adaptive fusion between spatial and spectral branches is achieved for discriminative information integrity. Extensive experiments conducted on four HSI datasets, i.e., Indian Pines, Pavia University, Houston 2013, and Houston 2018, demonstrate that the proposed HMSSF outperforms conventional classification methods as well as state-of-the-art CNN-based and transformer-based HSI methods. Future work will focus on extending the multiscale learning to diverse hyperspectral classification scenarios to enhance generalization capability, including exploration of extreme settings with limited training samples.

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